

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java Set Interface using HashSet

Objective / Aim

To implement the Set interface using "HashSet" and demonstrate adding, removing, searching, checking size, and clearing elements.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

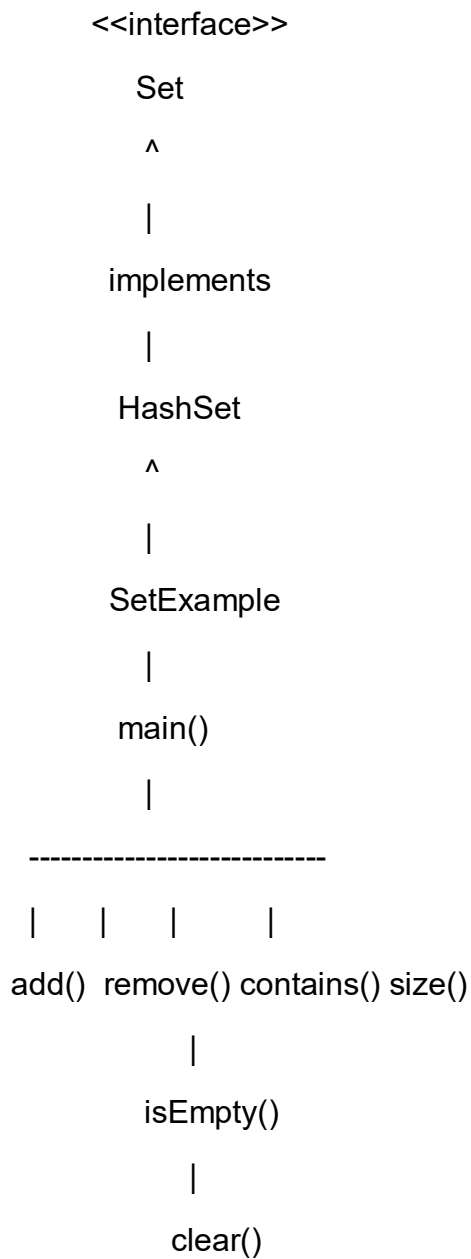
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "Set" using "HashSet".
2. Add elements using "add()".
3. Add a duplicate element and observe that it is not stored.
4. Remove an element using "remove()".
5. Check an element using "contains()".

6. Find the number of elements using "size()".
 7. Check whether the set is empty using "isEmpty()".
 8. Remove all elements using "clear()".
 9. End the program.
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UML Class Diagram



Source Code

```
package collectionframework;

import java.util.Set;
import java.util.HashSet;

public class SetExample {
    public static void main(String[] args) {

        Set<String> set = new HashSet<>();

        // add()
        set.add("Apple");
        set.add("Banana");
        set.add("Mango");
        set.add("Apple");    // Duplicate

        System.out.println("Set: " + set);

        // remove()
        set.remove("Banana");
        System.out.println("After remove: " + set);

        // contains()
        System.out.println("Contains Mango: " + set.contains("Mango"));

        // size()
        System.out.println("Size: " + set.size());
    }
}
```

```
// isEmpty()

System.out.println("Is Empty: " + set.isEmpty());


// clear()

set.clear();

System.out.println("After clear: " + set);

}

}
```

Compilation & Execution Commands

Save the file as:

SetExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac SetExample.java
```

Run:

```
java collectionframework.SetExample
```

Output Screen

Note: "HashSet" does not guarantee insertion order, so the display order may vary.

Set: [Apple, Mango, Banana]

After remove: [Apple, Mango]

Contains Mango: true

Size: 2

Is Empty: false

After clear: []

Result

The Set interface was successfully implemented using "HashSet", and the "add()", "remove()", "contains()", "size()", "isEmpty()", and "clear()" operations were successfully demonstrated. Duplicate elements were not stored.